

Lab Report: Installing Kali Linux in Hyper-V

Name: Mubeen Khan

Lab Objective

The primary objective of this lab was to successfully create and configure a nested virtual machine (VM) running Kali Linux within Hyper-V.

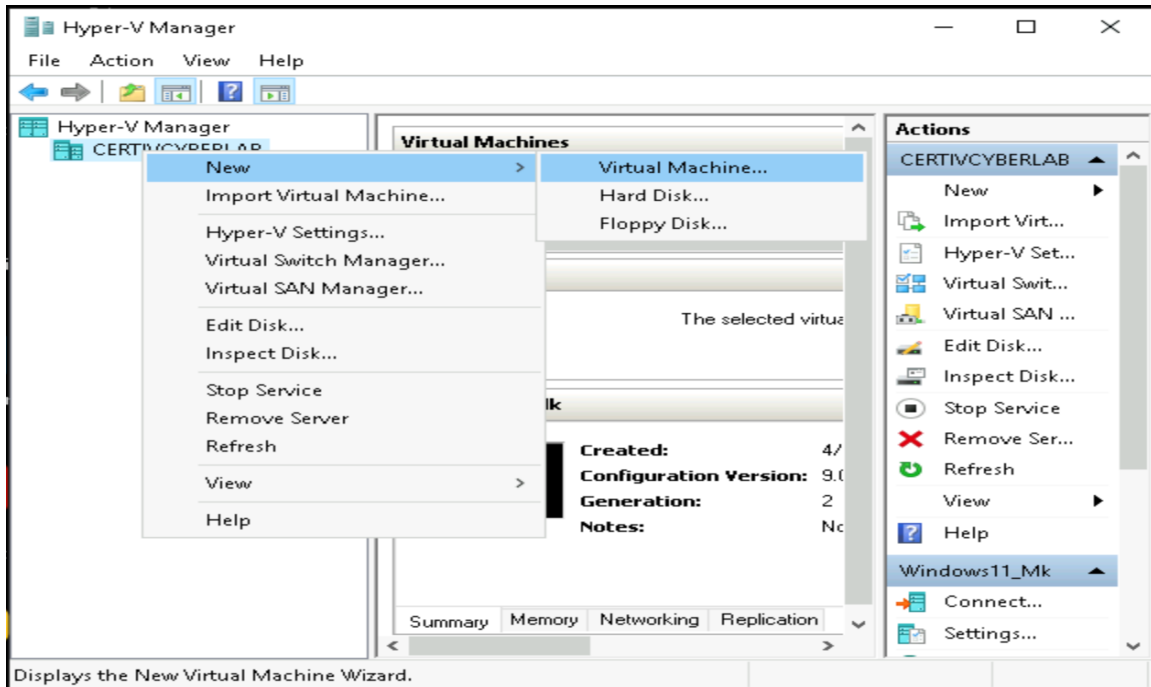
Tools / Environment Used

- Hyper-V Manager (Windows)
- Kali Linux VHDX (pre-existing virtual hard disk)
- Cert_IV_Cyber_Virtual_Switch (custom virtual switch)
- PowerShell (for enabling Enhanced Session Mode)
- Firefox (within Kali Linux)

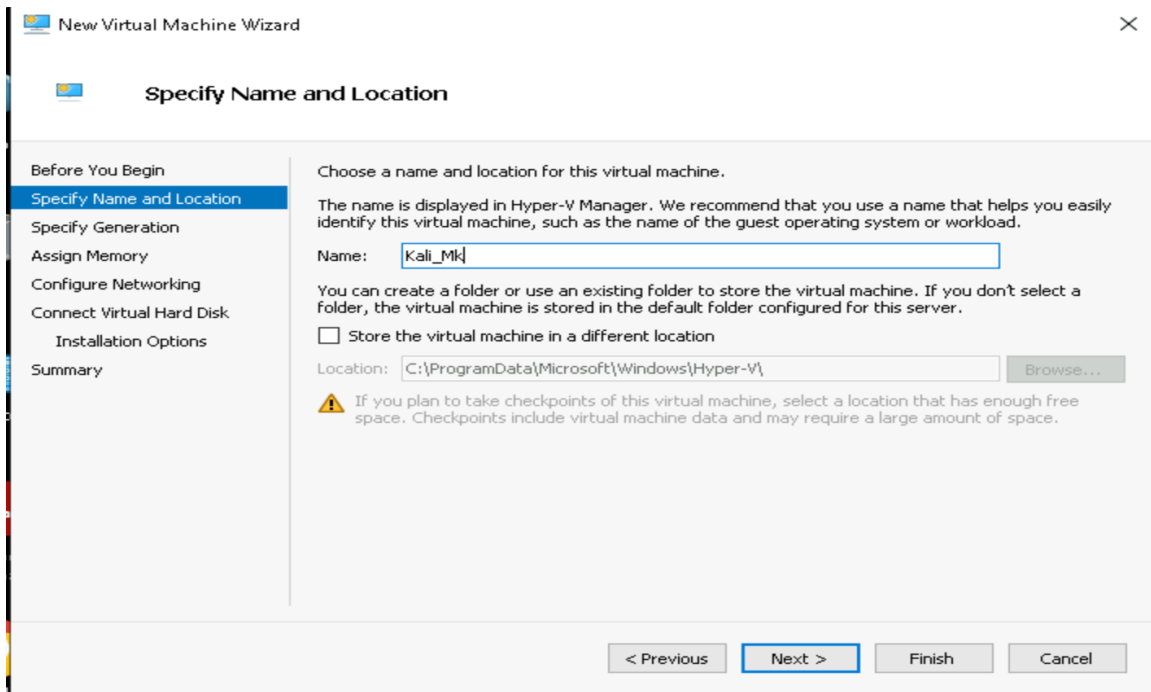
Task 1: Creating a Nested VM using Existing ISO

Step 1 – Creating the Virtual Machine

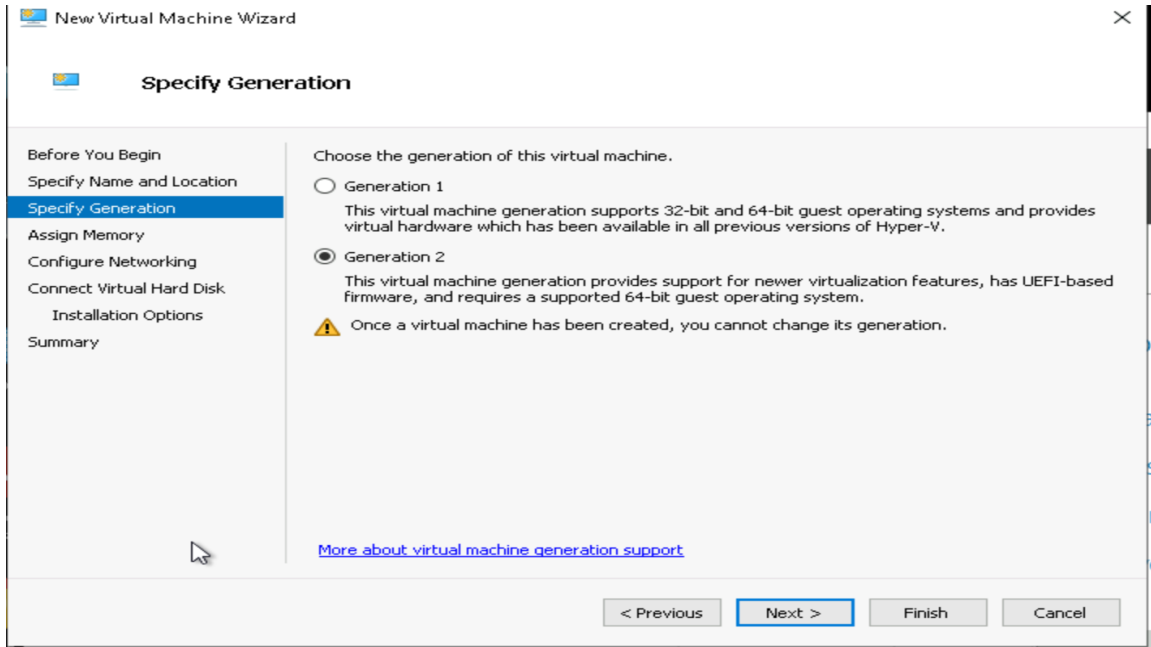
- Opened Hyper-V Manager from the desktop or Windows search.
- Right-clicked the server name → New → Virtual Machine.



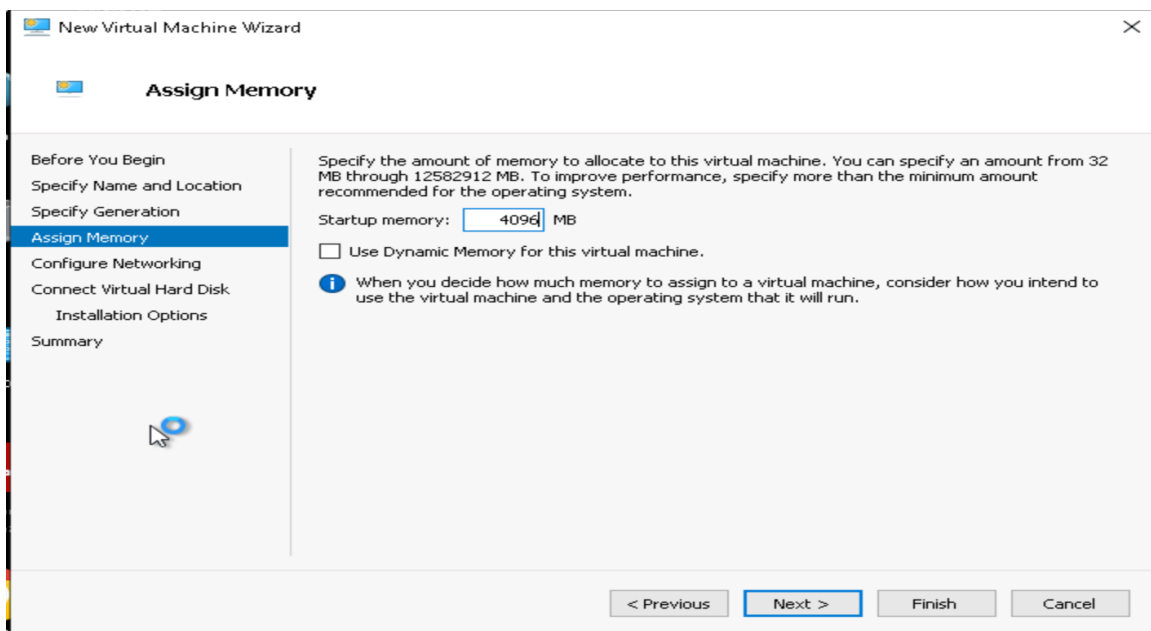
- Named the VM using the convention: Kali_[YourInitials].



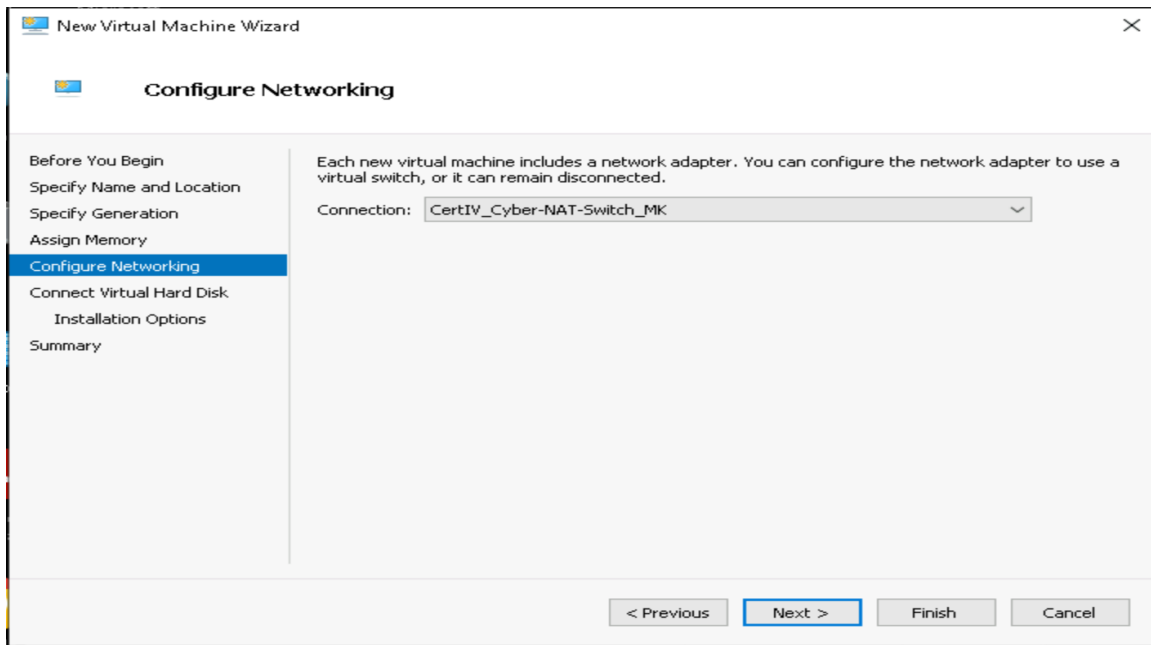
- Selected Generation 2.



- Assigned 4096 MB startup memory and disabled Dynamic Memory.



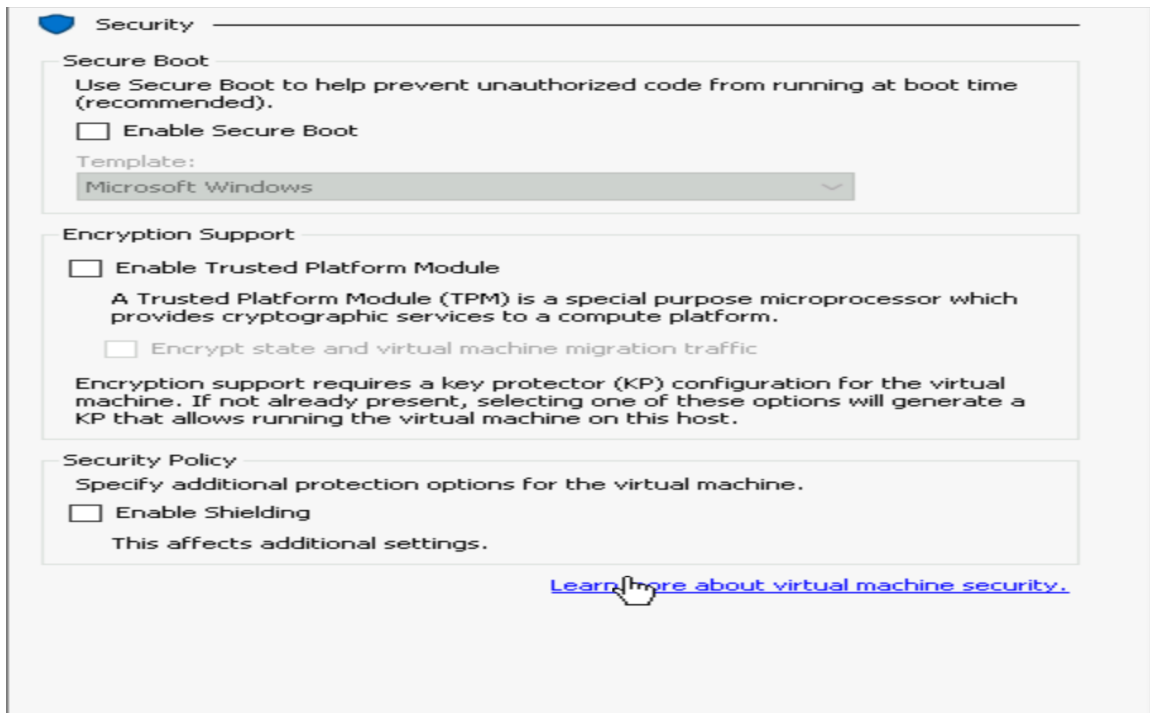
- Selected the virtual switch.



- Chose "Use an existing virtual disk" and selected Kali.vhdx.
- Completed the wizard.

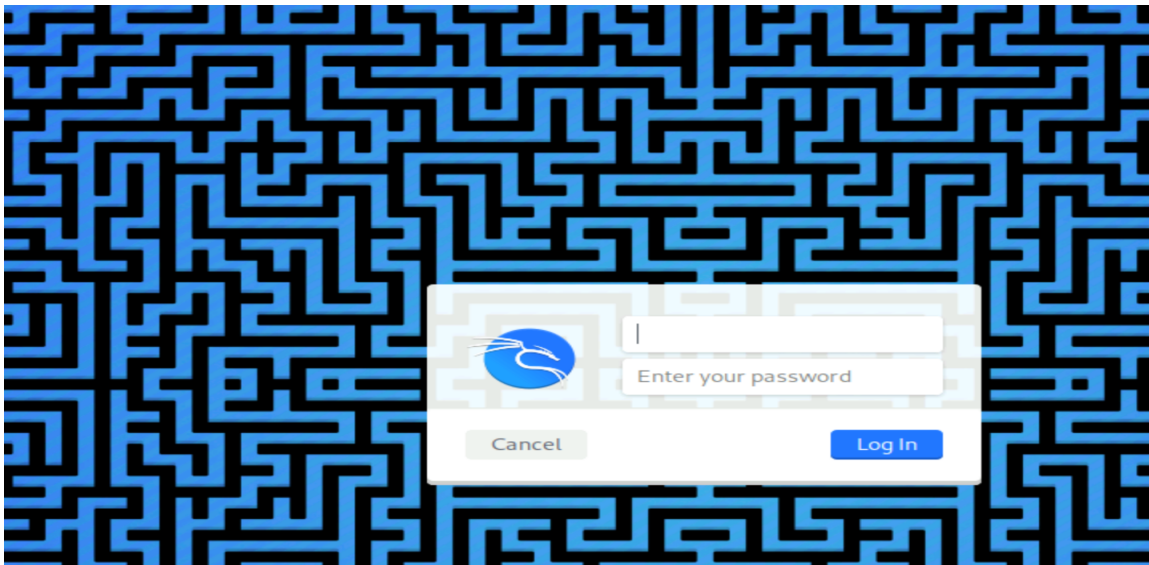
Step 2 – Adjusting VM Settings

- Increased virtual processors to 4.
- Disabled Secure Boot.
- Enabled Guest Services.



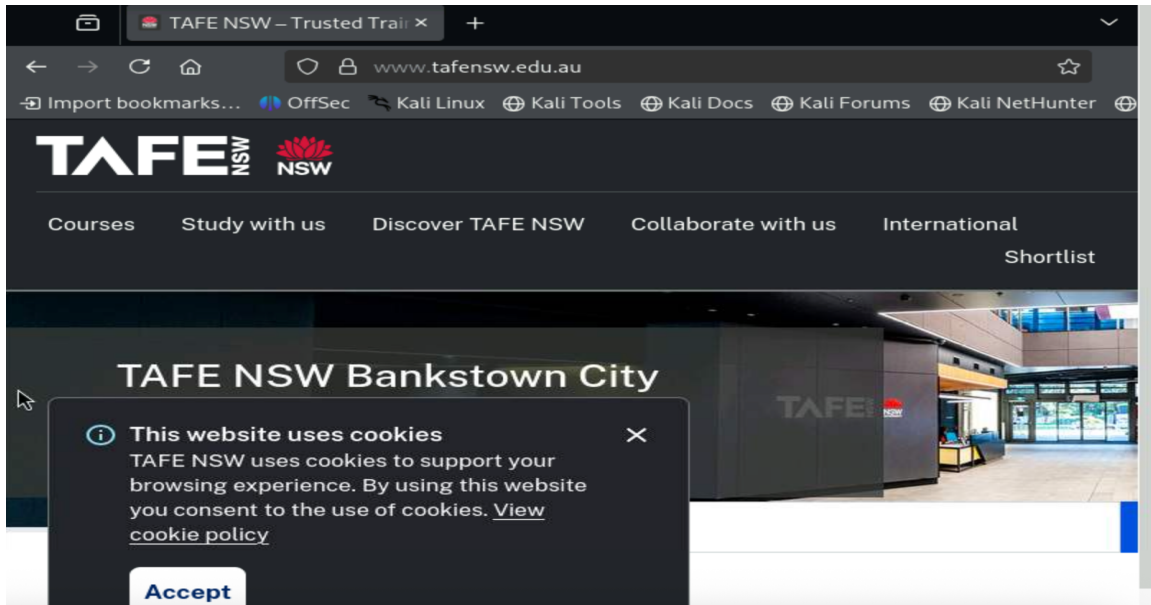
Task 2: Connecting to the Nested VM

- Connected to the VM and started it.
- Waited for Kali to boot.
- Logged in with username: student, password: student.



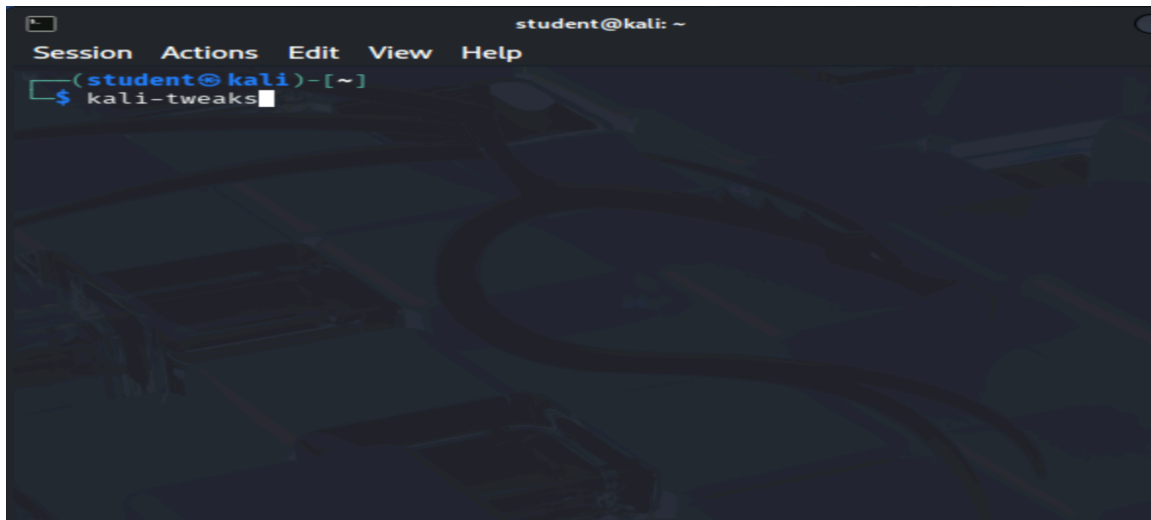
Task 3: Testing Internet Connectivity

- Opened Firefox.
- Navigated to www.tafensw.edu.au.
- Confirmed the site loaded successfully.

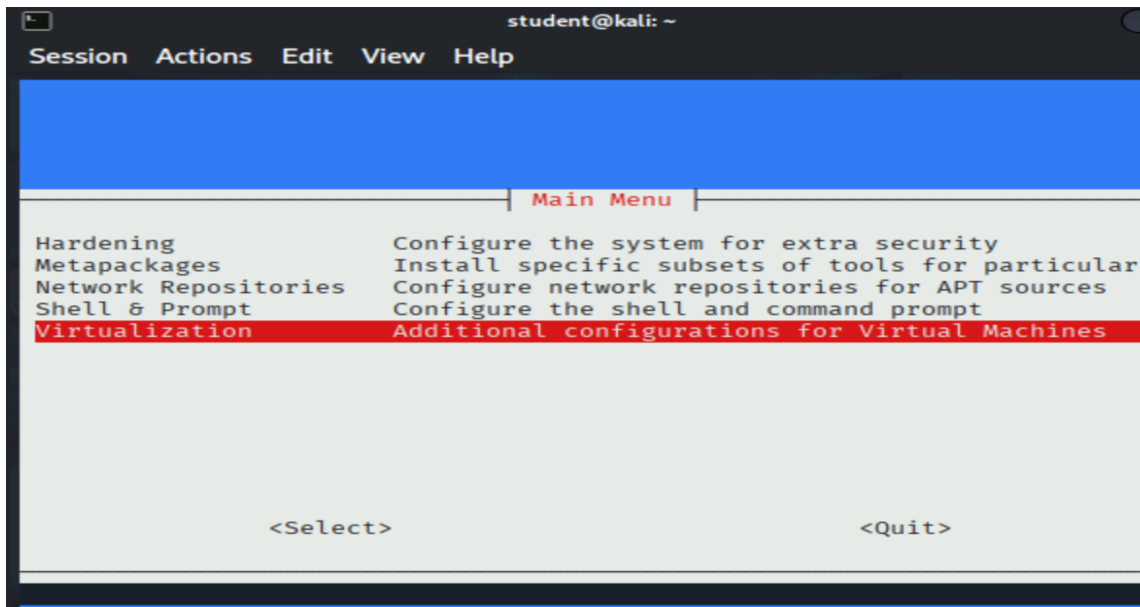


Task 4: Enabling Enhanced Session Mode

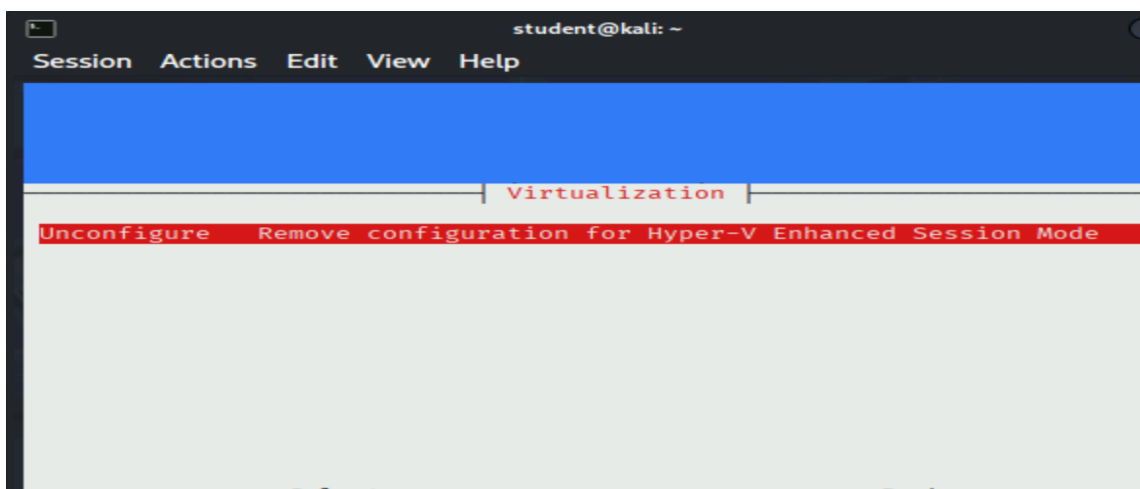
- Opened Terminal Emulator.
- Ran command: kali-tweaks



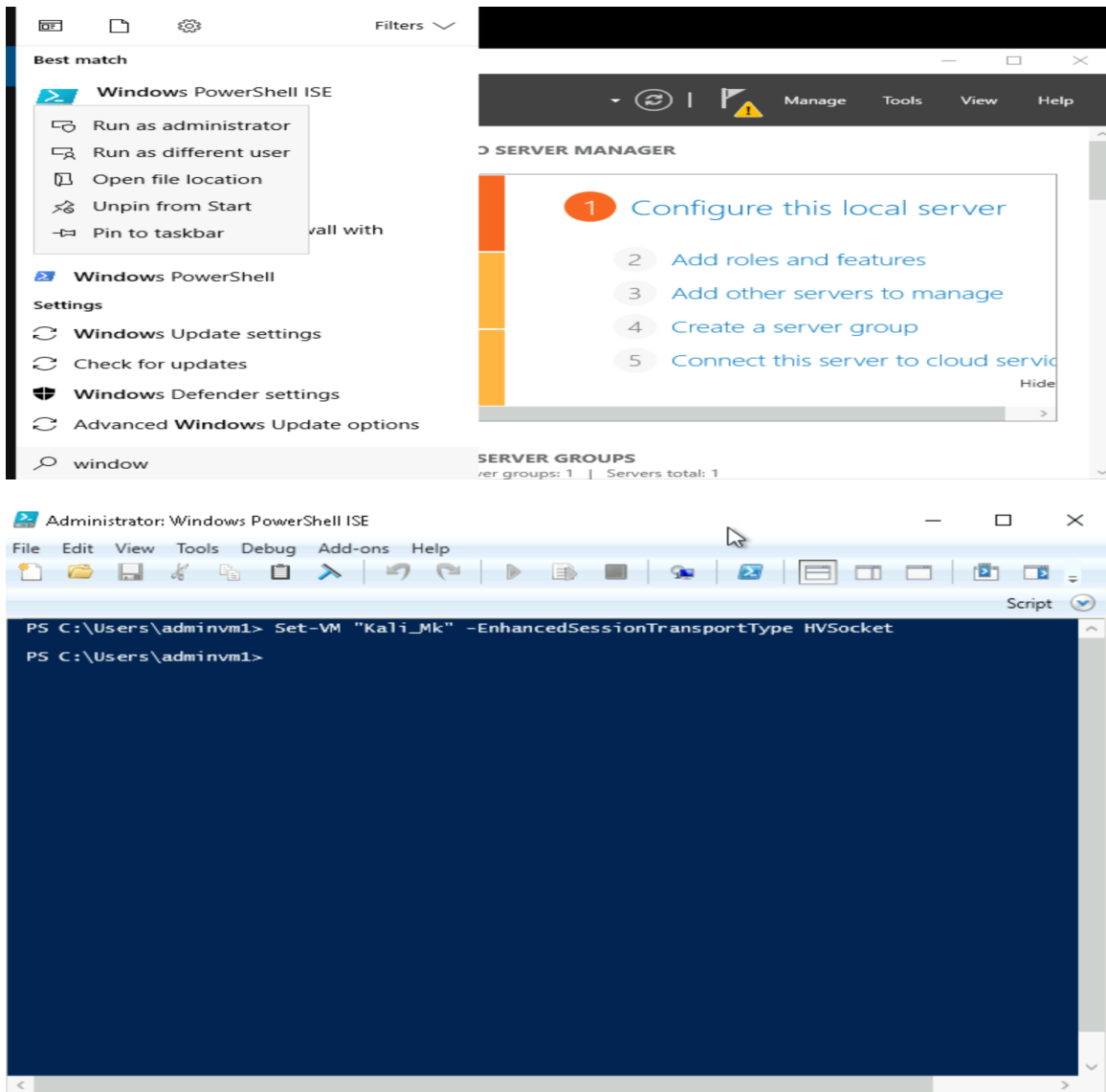
- Navigated to Virtualization settings.



- Entered password when prompted.



- Shut down the VM.
- Ran PowerShell command:
Set-VM "Kali_[YourInitials]" -EnhancedSessionTransportType HVSocket



Verification / Completion Checklist

- ✓ Kali VM created correctly
- ✓ RAM configured properly
- ✓ Virtual switch selected
- ✓ VHDX attached
- ✓ Processors updated
- ✓ Secure Boot disabled
- ✓ Guest Services enabled
- ✓ Login successful
- ✓ Internet connectivity verified
- ✓ Enhanced Session Mode enabled

Conclusion

In this lab, I successfully installed and configured a nested Kali Linux VM in Hyper-V, verified network connectivity, and enabled Enhanced Session Mode. These skills are essential for cybersecurity labs involving virtual environments. I confirm that all work is my own.